Luke Caffarey

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Mechanical engineering student (GPA 3.889) with hands-on experience across fusion research and utility-scale wind, now building custom rocket avionics and remote telemetry hardware. Drawn to flight hardware that has to work the first time.

EDUCATION

North Carolina State University *Raleigh, NC*

B.S. Mechanical Engineering | Expected May 2028 | GPA: 3.889

- Relevant tools and coursework: SolidWorks, MATLAB, Python, thermodynamics, fluid mechanics, mechanics of materials.

EXPERIENCE

GE Vernova, Wind Engineer Intern *May 2026 to Present | Greenville, SC*

- Support engineering on utility-scale wind turbine hardware within GE Vernova's onshore wind business.
- Automated a weekly safety-critical reporting workflow in Python and SQL, replacing a manual multi-step process.
- Apply CAD and mechanical analysis to hardware that must operate reliably in the field for decades.

NC State Mechanical Engineering, Senior Design Teaching Assistant *Aug 2026 to Present | Raleigh, NC*

- Support senior capstone design teams through design reviews, prototyping guidance, and documentation feedback.

FPAC Laboratory, NC State, Undergraduate Researcher *Dec 2024 to Apr 2025 | Raleigh, NC*

- Analyzed spectroscopy and video data from KSTAR fusion reactor experiments to support reactor wall stability.
- Built Python workflows for time-series signal processing, using adaptive Gaussian smoothing to improve peak detection.

PROJECTS

High-Power Rocket Avionics and Remote Ground Station (in progress) *2026*

- Designing a custom flight computer (Teensy 4.1, IMU, barometer, GPS) that logs 100 Hz flight data and streams live telemetry over a 915 MHz LoRa radio link.
- Building a Raspberry Pi ground station serving a real-time web dashboard for altitude, velocity, and recovery tracking, remotely accessible from anywhere over a Tailscale mesh VPN.
- Flight-testing the full telemetry stack on a quadcopter to shake out radio dropouts, range limits, and antenna placement before it flies on a high-power rocket.

LEADERSHIP AND INVOLVEMENT

Windpack Club, NC State, President *Aug 2025 to Present*

- Founded a 10-member team to compete in the Collegiate Wind Competition and led it to its first competition appearance in 2026.
- Lead design, fabrication, and testing of scaled wind turbine systems and sustainability proposals.

ADDITIONAL EXPERIENCE

Annelore's German Bakery, Assistant *Aug 2025 to Present | Cary, NC*

- Support weekend operations (16 hrs/week): baking assistance, order taking, and food prep.

Wake County Government, EMS Division, EMS Cadet *Jan 2022 to May 2024 | Raleigh, NC*

- Shadowed paramedics in 911 emergency response, assisting with patient care in high-pressure conditions.

SKILLS

CAD and Design: SolidWorks, design for manufacturing, 3D printing
Programming and Data: Python (signal processing, data analysis), MATLAB, SQL, C++ (embedded)
Hardware: Fabrication, soldering, sensor integration (IMU, barometer, GPS), LoRa radio telemetry
Systems: Raspberry Pi and Linux, Git, remote networking (SSH, Tailscale), thermodynamics, fluid mechanics